Project Proposal

Gridiron Intelligence: High School to College Player Projection System

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Problem Statement

College football recruiting is largely dependent on subjective scouting assessments, fragmented data sources, and star-based rankings, which frequently fail to effectively predict long-term player success. While there are large amounts of quantitative performance data (e.g., high school statistics, combine metrics) and qualitative scouting reports available, there is no consistent system that combines these inputs to produce an explainable, data-backed projection of a player's future collegiate success. Additionally, existing AI tools struggle to strike a balance between numerical reasoning and domain-specific language, producing results that are either statistically weak or insufficient for real-world scouting scenarios.

Target Users

- **College football recruiting analysts and support staff** seeking data-driven insights
- **High school football coaches and athletes** interested in realistic collegiate projections
- **Sports data analysts and researchers** studying talent development and outcomes
- **Media and scouting services** looking for scalable evaluation tools

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Proposed Solution

This project introduces **Gridiron Intelligence**, an AI-powered recruitment assistant that creates professional-style "Scouting Cards" for high school football players. This system combines machine learning with a domain-specific Retrieval-Augmented Generation (RAG) chatbot to generate **quantitative success potential** as well as **natural-language scouting** reports based on previous data.

To address the known limitation that large language models perform poorly at numerical reasoning, the system will be designed with two complementary engines:

1. A **Quantitative Prediction Engine** responsible for statistical modeling
2. A **Scouting Narrative Engine** responsible for qualitative analysis and explanation

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Key Features

- Predictive modeling of collegiate success tiers (e.g., Power 5 starter, NFL draft pick)
- Domain-specific scouting language generation mimicking professional evaluators
- Explainable projections backed by historical player comparisons
- Integration of structured performance data and unstructured scouting reports
- Modular architecture enabling scalability and experimentation

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Technical Stack (APIs, Frameworks, Databases)

Programming Language

- Python

Machine Learning & Modeling

- Scikit-learn
- XGBoost or Random Forests

Large Language Model & RAG Framework

- LangChain or LlamaIndex
- Base LLM: Gemini 3.0 Flash

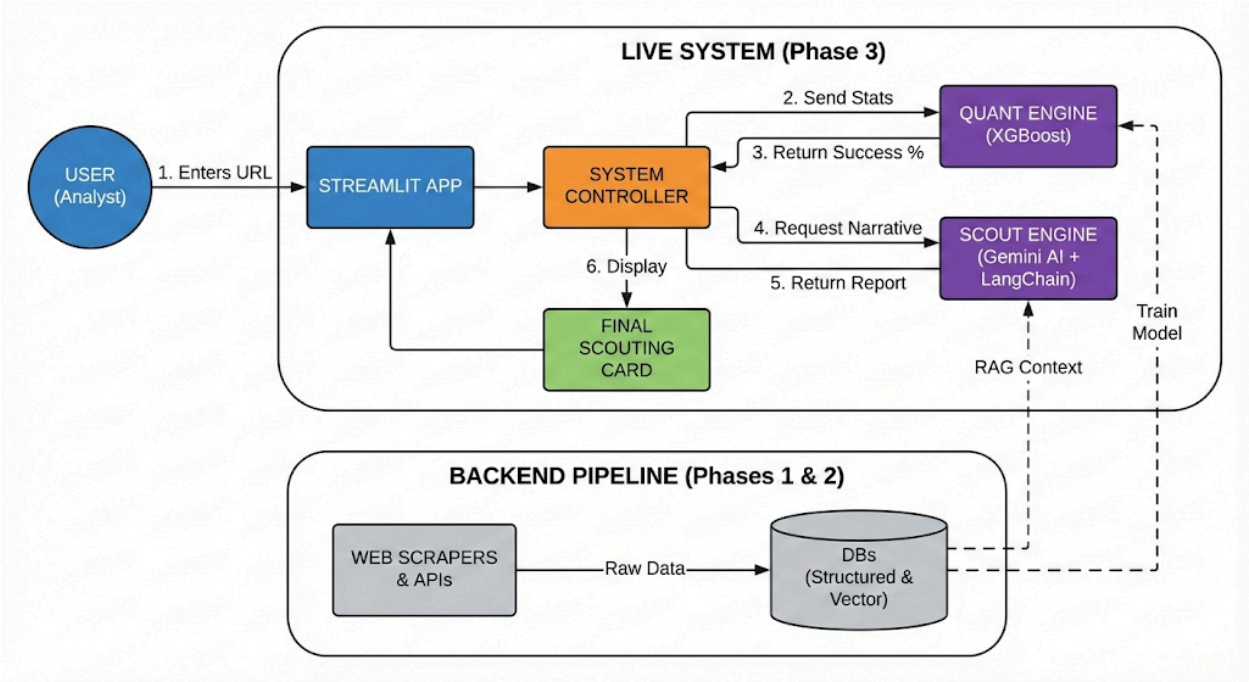
Data Storage & Retrieval

- Vector Database: ChromaDB (local development) or Pinecone
- Structured data storage: CSV / Parquet files

Data Sources (Publicly Available)

- High School Performance Data: MaxPreps
- Scouting Reports & Ratings: 247Sports, Rivals (text-based analysis only)
- Outcome Data: College Football Reference

System Architecture Diagram



Success Criteria and Evaluation Metrics

Model Performance Metrics

- Classification accuracy for success tier prediction
- Precision and recall for top-tier outcomes (Power 5 starter, NFL draft pick)
- Cross-validation performance on historical classes

System-Level Evaluation

- Consistency between quantitative predictions and narrative explanations
- Quality and realism of scouting language (manual qualitative review)
- Ability to retrieve meaningful historical player comparisons

Acceptance Criteria

- Successful end-to-end pipeline execution on historical and current data
- Generation of complete scouting cards for test players
- Demonstrated alignment with course concepts (ML modeling, RAG, system design)